PROJECT CASE STUDY

Python Point-of-Sale Desktop App

Inventory, Receipts, Sales Logs & Windows Packaging

Category	Status	Year
Software Engineering	Packaged application	2026

A complete desktop POS workflow using an Excel product database, barcode lookup, stock updates, PDF receipts, daily sales logs, and a PyInstaller Windows executable.

EXE	PDF	Excel
Windows package	receipt output	inventory + logs

Context and problem

A small retail shop needs a simple offline system for products, billing, stock control, customer change, receipts, and daily records without operating a complex server.

Strategy and implementation

- Created a barcode-based shopping cart and quantity workflow.
- Loaded and updated inventory through Excel.
- Generated formatted PDF receipts with ReportLab.
- Logged daily sales to Excel and packaged the full app with PyInstaller.

My role

Application logic, data access, cart behavior, stock validation, receipt design, logging, and Windows packaging.

Technology stack

Python | Tkinter | Pandas | Excel | ReportLab | PyInstaller

Results and evidence

- Produced a directly runnable Windows executable.
- Implemented stock-safe checkout, discounts, cash received, and change.
- Generated persistent PDF receipts and daily Excel sales records.

Key takeaways

- Deployment includes packaging, paths, and file permissions - not only application code.
- Offline-first software can be the right architecture for a small business.

Portfolio: github.com/awai1s | awaisraza5424@gmail.com